

```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("DataSet.csv")
df.head()

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 17880,\n  \"fields\": [\n    {\n      \"column\": \"title\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 11231,\n        \"samples\": [\n          \"Environmental Lab Technician\",\n          \"Junior Designer\",\n          \"Case Handler\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"location\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 3105,\n        \"samples\": [\n          \"US, VA, Arlington\",\n          \"US, AL, Dothan\",\n          \"US, CA, Long Beach\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"department\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 1337,\n        \"samples\": [\n          \"Finance team\",\n          \"Product Ops\",\n          \"Engineering - Hardware\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"salary_range\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 874,\n        \"samples\": [\n          \"60000-130000\",\n          \"30000-450000\",\n          \"120000-130000\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"company_profile\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 1710,\n        \"samples\": [\n          \"<p>We make tools for startups. Currently building <a rel=\\\"nofollow\\\" class=\\\"external\\\">Hire my Friend</a>, <a rel=\\\"nofollow\\\" class=\\\"external\\\">Wrangler</a>, <a rel=\\\"nofollow\\\" class=\\\"external\\\">Attending</a>, <a rel=\\\"nofollow\\\" class=\\\"external\\\">Linkydink</a>. Based in London and proud of it!</p>\",\n          \"<p>AbZorba is a world leading mobile casino games studio based in Athens that recently smashed through 2 million gamer installs, reached #1 US iOS Casino Games and has #1 Blackjack Android worldwide.</p>\",\n          \"<p>Internet Engine is a dynamic and diverse Internet marketing company specializing in Search Engine Optimization (SEO), Search Engine Marketing (SEM), rapid website deployment and lead generation for Real Estate professionals. We have developed a successful unified marketing strategy that generates targeted traffic to websites which provides hundreds of leads each week in several lucrative Real Estate markets.</p>\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}

```

```

\ "description\"," \n \ "properties\":" { \n \ "dtype\":"
\ "string\"," \n \ "num_unique_values\":" 15095, \n
\ "samples\":" [ \n \ "<p>Luxury beauty & fragrance
consultants needed!</p>\r\n<p>Pure Placements\\u00a0are a specialist
luxury retail recruitment agency, specialising is temporary &
permanent beauty, fragrance, fashion & retail staff.\\u00a0We are
recruiting now for beauty & fragrance consultants to work within
department stores across the country promoting luxury products such as
make up, skin care & perfume, must come from a beauty/retail
background,\\u00a0exceptional customer service skills are a must!\\
u00a0If you feel you have relevant experience and want to join our
fantastic team\\u00a0please email a copy of\\u00a0your CV with a
recent full length or head shot\\u00a0photo to marie<a
href=\\\"mailto:#EMAIL_6028ae5388335a3013c32058435e158c80f0014f2796bd5
12ba7ab0c6b01cd26#\\\" rel=\\\"nofollow\\\"
class=\\\"external\\\">@#URL_2cf2adf737014de8b49f0776d9c8caala359f0bf4
2fa9910bb760d5475a2f1b9#</a></p>\", \n \ "<p>Temporary Service
Desk Analysts \\u00a0will perform various activities related to
managing general Desktop and Server information technology. This
includes, but is not limited to: Email (Exchange, others), Directory
Services (AD), Windows, Linux, Citrix, Vmware, mobile devices, e-mail
(Exchange/Office 365), and Windows Server applications.</p>\r\n
<p>This position is a direct hire by Virteva; no staffing agency is
involved.\\u00a0</p>\r\n<p><b>ESSENTIAL DUTIES AND
RESPONSIBILITIES:</b></p>\r\n<ul>\r\n<li>Taking initial telephone
inquiries and troubleshooting relatively simple hardware, software, or
network problems.</li>\r\n<li>Monitor incoming tickets and alerts in
the ServiceNow system and respond accordingly.</li>\r\n<li>Update
all tickets according to Service Level Agreements (SLA).</li>\r\n
<li>Working as an effective team member with the Virteva team,
partners and internal staff.</li>\r\n<li>Maintaining professional,
quality interactions with customers at all times.</li>\r\n
<li>Recognizing and escalating more difficult problems</li>\r\n
<li>Documenting customer impact, resolution steps, and appropriate
priority in accordance with policy and customer requirements.</li>\r
r\n<li>Other duties as assigned.\\u00a0</li>\r
r\n</ul><p><b>REQUIREMENTS:</b></p>\r\n<ul>\r\n<li>We require a
commitment until the project is complete, on a contract basis, 2-4
months, 40 hours a week, standard business hours.</li>\r\n
<li>Background check will be required.</li>\r\n</ul>\", \n
\ "<p>Our mission at MarketInvoice is to modernise the way by which
businesses finance their working capital and fund their growth. We are
seeking to bring much-needed innovation to the banking industry in
much the same way that online retailers disrupted the high
street.</p>\r\n<p>We offer a new kind of working capital finance,
built for savvy growing businesses. Our online platform gives clients
access to funds in outstanding individual invoices, otherwise tied up
for between 30 to 120 days. MarketInvoice is entirely different to
traditional factoring. Businesses can sign up online, sell an invoice

```

and draw down funds on the same day. There are no contracts, hidden fees or personal guarantees.

We launched in 2011 and already we've helped hundreds of growing companies overcome their fluctuating cash-flow. Over £320m has been funded since we started. Our clients typically use the finance we help them to raise to launch new products, hire more staff or expand internationally.

A global pool of sophisticated investors fund the invoices on the platform, and as of August 2013 the UK government began buying invoices as part of the British Business Bank initiative. We've had lots of praise in the press, both in the UK and internationally, and we're winning some big awards, including a 2014 Santander Breakthrough 50 award.

Based in central London, you will be joining a young, multi-talented fast-growing team passionate about using technology to help businesses access funding, and changing the world of bank finance.

The Role

As Director of Operations you will be responsible for the daily operations of MarketInvoice. You will ensure that our systems can securely and efficiently settle thousands of invoice funding transactions flowing through our platform. You will be part of product planning and management and will inform product decisions based on your knowledge of our current and future operational capabilities. You will report to the CEO and be the point person for implementation of our strategy and for design and management of supply chains within our business.

```

n      ],n      {"semantic_type": "\"",n
{"description": "\"\"n      }n      },n      {n      {"column":
"requirements",n      {"properties": {n      {"dtype":
"string",n      {"num_unique_values": 12119,n
"samples": [n      {"<ul>\\r\\n<li>Seasoned iOS engineer: armed
with strong Objective-C coding skills, you have solid hands-on
experience with the iOS core framework and libraries.</li>\\r\\
n<li>Good understanding of JavaScript, HTML(5) and browser
technologies.</li>\\r\\n<li>Solid design and prototyping
skills.</li>\\r\\n<li>Strong CS fundamentals - algorithms, data
structures, operating systems, networking.</li>\\r\\n<li>Impeccable
software development skills with a solid understanding of design
patterns and OO/functional programming paradigms.</li>\\r\\n<li>Proven
debugging and troubleshooting skills. You know how to use\\
u00a0debugging/profiling tools in Xcode or on the command line if
needed.</li>\\r\\n<li>Comfortable working in a test-driven development
environment.</li>\\r\\n<li>Experience working in a continuous
integration development cycle.</li>\\r\\n<li>Excellent written and
verbal communications skills at the business and technical
level.</li>\\r\\n<li>Strong work ethic, commitment and team
work.</li>\\r\\n<li>Fluent in writing well-documented code, explaining
methods in good English.</li>\\r\\n<li>A degree in computer science or
similar is a plus.</li>\\r\\n<li>Android/Java development experience
is a plus.</li>\\r\\n<li>We greatly value participation in Linux /
Open Source projects. We live and breathe Open Source.</li>\\r\\

```

nSelf-motivated, but can excel in a team-based environment. Strong initiative and enthusiasm.\\r\\nYou have to be eligible to work in the US.\\r\\n\",\\n \\<p>Position Qualifications</p>\\r\\n\\r\\nErgonomic certified or willing to take the on line certification class within the first 90 days of employment\\r\\nProven supervisory/leadership experience; Minimum of\\u00a03 years management experience in logistics, facilities or shipping/receiving required\\r\\nManagement experience in multi-service environment; minimum 3 years\\r\\nEffective customer relationship building skills\\r\\nEffective written and verbal communication skills; proven analytical skills\\r\\nExcellent planning, prioritization and organizational skills\\r\\nAbility to resolve issues without management supervision required\\r\\nProven experience in achieving financial goals related to revenue and profitability\\r\\nProficiency in email environments, MS Office Suite (Word, Excel required, PowerPoint preferred)\\r\\nMinimum of a HS diploma or equivalent required, undergraduate degree preferred\\r\\nAbility to administer company policies and procedures\\r\\nWillingness to submit to a pre-employment drug screening and criminal background check\\r\\nValid driver's license and clean driving record (less than 3 violations the past 3 years)\\r\\n\",\\n \\<p>Knowledge, Skills and Experience</p>\\r\\n\\r\\n\\u00a06 years of experience in product management preferably in the telecommunications services sector.\\r\\n\\u00a0Telecommunications domain knowledge \\u2013 must understand the mobile operator business and challenges\\r\\nB. Science in Technology or Business. MBA a plus.\\r\\nHands on experience in managing the development of an idea, creating the specifications, managing the internal reviews and market research.\\r\\n\\u00a0Ability to formulate requirements and experience working with research and development\\r\\n\\u00a0Experience with collaboration with a multi geography and multi-functional virtual team\\r\\nPrevious P&L responsibility\\u00a0\\r\\nPossess working knowledge of technologies applied to product implementation.\\r\\nIncrease account penetration and pipeline diversity with new solution development\\r\\nStrong understanding of competitors and ability to explain competitive differentiators\\r\\nDemonstrated ability and experience developing a product through its life cycle\\r\\nAbility to identify key differentiators, features and benefits, and be able to credibly articulate these to internal stakeholders, customers/prospects, analysts/media, etc\\r\\nProven leadership credentials with an emphasis on positive team spirit and building virtual teams\\r\\n\\u00a0Experience working with sales to help define/refine sales messages and ensure sales packages are developed and messages communicated to sales team\\r\\n<p>Personal Characteristics</p>\\r\\n

```

n<ul>\\r\\n<li>Able to perform under pressure and deliver results in a
demanding and fast-paced environment that requires fresh thinking and
innovation</li>\\r\\n<li>\\u00a0Strategic thinking and solution
oriented approach</li>\\r\\n<li>\\u00a0Excellent communication skills,
both written and oral, ability to communicate with CX levels and team
work skills</li>\\r\\n<li>\\u00a0 Must possess an entrepreneurial mind
set and spirit</li>\\r\\n<li>\\u00a0Need to be a focused, proactive
self-starter who can operate in a constantly evolving team environment
and have the ability to work effectively with cross-functional and
remote teams.</li>\\r\\n</ul>\"\\n
    ],\\n
    \"semantic_type\": \"\",\\n        \"description\": \"\"\\n
    }\\n    },\\n    {\\n        \"column\": \"benefits\",\\n        \"properties\":
    {\\n            \"dtype\": \"category\",\\n            \"num_unique_values\":
    6510,\\n            \"samples\": [\\n                \"<p>Level 2 qualification
after one year and a permanent role thereafter.</p>\\r\\n<p>Great job
satisfaction</p>\\r\\n<p></p>\",\\n                \"<p><b>What we
offer:</b></p>\\r\\n<ul>\\r\\n<li>Great team work environment</li>\\r\\n<li>Competitive salary</li>\\r\\n<li>Paid holidays, vacation and
personal time off</li>\\r\\n<li>Opportunities for professional
development and advancement</li>\\r\\n</ul>\",\\n                \"<ul>\\r\\n
<li>Competitive compensation with uncapped commissions</li>\\r\\n
<li>Collaborative, open culture that encourages coaching and
development</li>\\r\\n<li>Full benefits including access to Weston
Fitness (located in our building)</li>\\r\\n<li>Friday lunches and
company outings</li>\\r\\n<li>Fully stocked kitchen and pantry</li>\\r\\n
<li>Opportunity to quickly grow into new and exciting roles in
fast growing technology company</li>\\r\\n</ul><p>\\u00a0</p>\"\\n
    ],\\n        \"semantic_type\": \"\",\\n        \"description\": \"\"\\n
    }\\n    },\\n    {\\n        \"column\": \"telecommuting\",\\n
    \"properties\": {\\n            \"dtype\": \"category\",\\n
    \"num_unique_values\": 2,\\n            \"samples\": [\\n                \"t\",\\n
    \"f\"\\n            ],\\n        \"semantic_type\": \"\",\\n
    \"description\": \"\"\\n        }\\n    },\\n    {\\n        \"column\":
    \"has_company_logo\",\\n        \"properties\": {\\n            \"dtype\":
    \"category\",\\n            \"num_unique_values\": 2,\\n            \"samples\":
    [\\n                \"f\",\\n                \"t\"\\n            ],\\n
    \"semantic_type\": \"\",\\n            \"description\": \"\"\\n        }\\n
    },\\n    {\\n        \"column\": \"has_questions\",\\n
    \"properties\": {\\n            \"dtype\": \"category\",\\n
    \"num_unique_values\": 2,\\n            \"samples\": [\\n                \"t\",\\n
    \"f\"\\n            ],\\n        \"semantic_type\": \"\",\\n
    \"description\": \"\"\\n        }\\n    },\\n    {\\n        \"column\":
    \"employment_type\",\\n        \"properties\": {\\n            \"dtype\":
    \"category\",\\n            \"num_unique_values\": 5,\\n            \"samples\":
    [\\n                \"Full-time\",\\n                \"Temporary\"\\n            ],\\n
    \"semantic_type\": \"\",\\n            \"description\": \"\"\\n        }\\n
    },\\n    {\\n        \"column\": \"required_experience\",\\n
    \"properties\": {\\n            \"dtype\": \"category\",\\n
    \"num_unique_values\": 7,\\n            \"samples\": [\\n

```



```

\ "Internship\ ",\n          \ "Not Applicable\ "\n          ],\n
\ "semantic_type\ ": \ "\",\n          \ "description\ ": \ "\",\n          }\n
n      },\n      {\n          \ "column\ ": \ "required_education\ ",\n
\ "properties\ ": {\n          \ "dtype\ ": \ "category\ ",\n
\ "num_unique_values\ ": 13,\n          \ "samples\ ": [\n
\ "Vocational - Degree\ ",\n          \ "Doctorate\ "\n          ],\n
\ "semantic_type\ ": \ "\",\n          \ "description\ ": \ "\",\n          }\n
n      },\n      {\n          \ "column\ ": \ "industry\ ",\n          \ "properties\ ":
{\n          \ "dtype\ ": \ "category\ ",\n          \ "num_unique_values\ ":
131,\n          \ "samples\ ": [\n          \ "Publishing\ ",\n
\ "Music\ "\n          ],\n          \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\",\n          }\n      },\n      {\n          \ "column\ ":
\ "function\ ",\n          \ "properties\ ": {\n          \ "dtype\ ":
\ "category\ ",\n          \ "num_unique_values\ ": 37,\n
\ "samples\ ": [\n          \ "Human Resources\ ",\n          \ "Business
Development\ "\n          ],\n          \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\",\n          }\n      },\n      {\n          \ "column\ ":
\ "fraudulent\ ",\n          \ "properties\ ": {\n          \ "dtype\ ":
\ "category\ ",\n          \ "num_unique_values\ ": 2,\n          \ "samples\ ":
[\n          \ "t\ ",\n          \ "f\ "\n          ],\n          \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\",\n          }\n      },\n      {\n          \ "column\ ": \ "in_balanced_dataset\ ",\n
\ "properties\ ": {\n          \ "dtype\ ": \ "category\ ",\n
\ "num_unique_values\ ": 2,\n          \ "samples\ ": [\n          \ "t\ ",\n
\ "f\ "\n          ],\n          \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\",\n          }\n      }\n      ]\n
n}","type":"dataframe","variable_name":"df"}

```

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17880 entries, 0 to 17879
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   title                 17880 non-null  object
 1   location              17534 non-null  object
 2   department            6333 non-null   object
 3   salary_range          2868 non-null   object
 4   company_profile       14572 non-null   object
 5   description           17880 non-null   object
 6   requirements          15191 non-null   object
 7   benefits              10684 non-null   object
 8   telecommuting         17880 non-null   object
 9   has_company_logo      17880 non-null   object
10   has_questions         17880 non-null   object
11   employment_type       14409 non-null   object
12   required_experience    10830 non-null   object
13   required_education    9775 non-null    object
14   industry              12977 non-null   object

```

```

15 function          11425 non-null object
16 fraudulent        17880 non-null object
17 in_balanced_dataset 17880 non-null object

```

```
dtypes: object(18)
```

```
memory usage: 2.5+ MB
```

```

df = df[['telecommuting',
        'has_company_logo',
        'has_questions',
        'employment_type',
        'required_experience',
        'fraudulent']]

```

```
df.head()
```

```

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 17880,\n  \"fields\": [\n    {\n      \"column\": \"telecommuting\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"t\",\n          \"f\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"has_company_logo\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"t\",\n          \"f\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"has_questions\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"t\",\n          \"f\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"employment_type\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \"Full-time\",\n          \"Temporary\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"required_experience\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 7,\n        \"samples\": [\n          \"Internship\",\n          \"Not Applicable\",\n          \"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"fraudulent\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"f\",\n          \"t\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    ]\n  }},\n  \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```
df = df.dropna()
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 10597 entries, 0 to 17879
```

```
Data columns (total 6 columns):
```

#	Column	Non-Null Count	Dtype
0	telecommuting	10597 non-null	object
1	has_company_logo	10597 non-null	object
2	has_questions	10597 non-null	object
3	employment_type	10597 non-null	object
4	required_experience	10597 non-null	object
5	fraudulent	10597 non-null	object

```
dtypes: object(6)
```

```
memory usage: 579.5+ KB
```

```
df['telecommuting'] = df['telecommuting'].map({'f':0, 't':1})
df['has_company_logo'] = df['has_company_logo'].map({'f':0, 't':1})
df['has_questions'] = df['has_questions'].map({'f':0, 't':1})
df['fraudulent'] = df['fraudulent'].map({'f':0, 't':1})
```

```
df.head()
```

```
{
  "summary": {
    "name": "df",
    "rows": 10597,
    "fields": [
      {
        "column": "telecommuting",
        "properties": {
          "dtype": "number",
          "std": 0,
          "min": 0,
          "max": 1,
          "num_unique_values": 2,
          "samples": [1, 0],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "has_company_logo",
        "properties": {
          "dtype": "number",
          "std": 0,
          "min": 0,
          "max": 1,
          "num_unique_values": 2,
          "samples": [1, 0],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "employment_type",
        "properties": {
          "dtype": "category",
          "num_unique_values": 5,
          "samples": ["Full-time", "Temporary"],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "required_experience",
        "properties": {
          "dtype": "category",
          "num_unique_values": 7,
          "samples": ["Internship", "Not Applicable"],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "fraudulent",
        "properties": {
          "dtype": "number",
          "std": 0,
          "min": 0,
          "max": 1
        }
      }
    ]
  }
}
```



```

{"num_unique_values": 2, "samples": [0, 1], "semantic_type": "\"", "description": "\"\""}
{"type": "dataframe", "variable_name": "df"}

```

```
from sklearn.preprocessing import LabelEncoder
```

```
le1 = LabelEncoder()
```

```
le2 = LabelEncoder()
```

```
df['employment_type'] = le1.fit_transform(df['employment_type'])
```

```
df['required_experience'] =
```

```
le2.fit_transform(df['required_experience'])
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 10597 entries, 0 to 17879
```

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	telecommuting	10597 non-null	int64
1	has_company_logo	10597 non-null	int64
2	has_questions	10597 non-null	int64
3	employment_type	10597 non-null	int64
4	required_experience	10597 non-null	int64
5	fraudulent	10597 non-null	int64

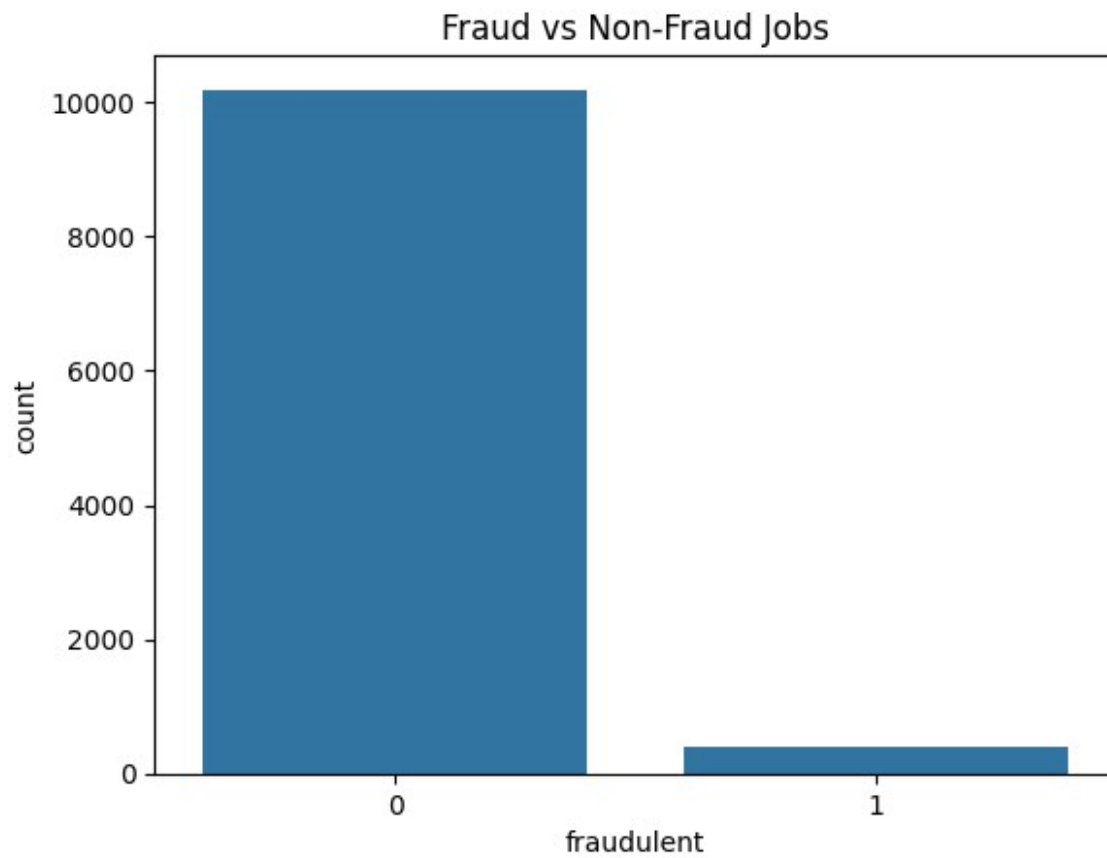
```
dtypes: int64(6)
```

```
memory usage: 579.5 KB
```

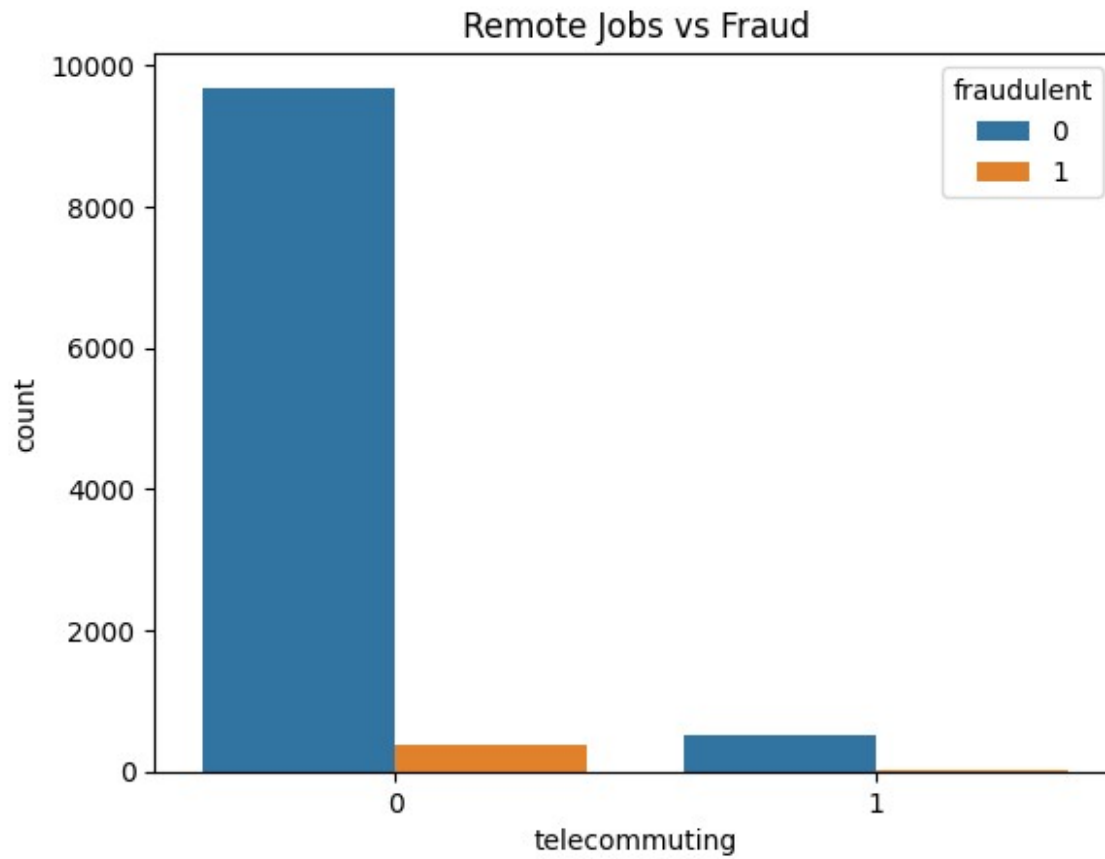
```
sns.countplot(x='fraudulent', data=df)
```

```
plt.title("Fraud vs Non-Fraud Jobs")
```

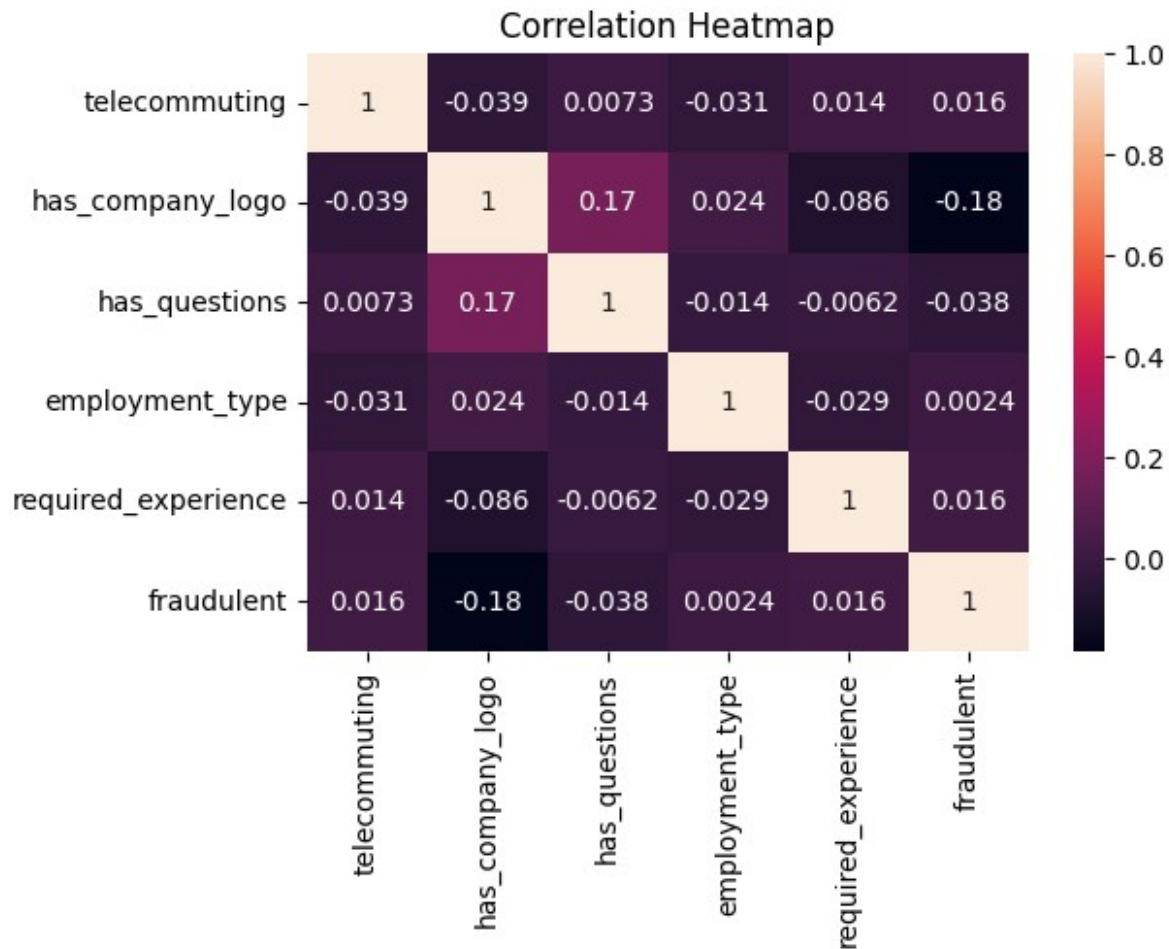
```
plt.show()
```



```
sns.countplot(x='telecommuting', hue='fraudulent', data=df)  
plt.title("Remote Jobs vs Fraud")  
plt.show()
```



```
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



```

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix

# Create model with class balancing
model = LogisticRegression(class_weight='balanced')

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Results
print("Classification Report:\n")
print(classification_report(y_test, y_pred))

print("Confusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))

Classification Report:

```

	precision	recall	f1-score	support
0	0.98	0.85	0.91	2034
1	0.14	0.55	0.22	86
accuracy			0.84	2120
macro avg	0.56	0.70	0.57	2120
weighted avg	0.94	0.84	0.88	2120

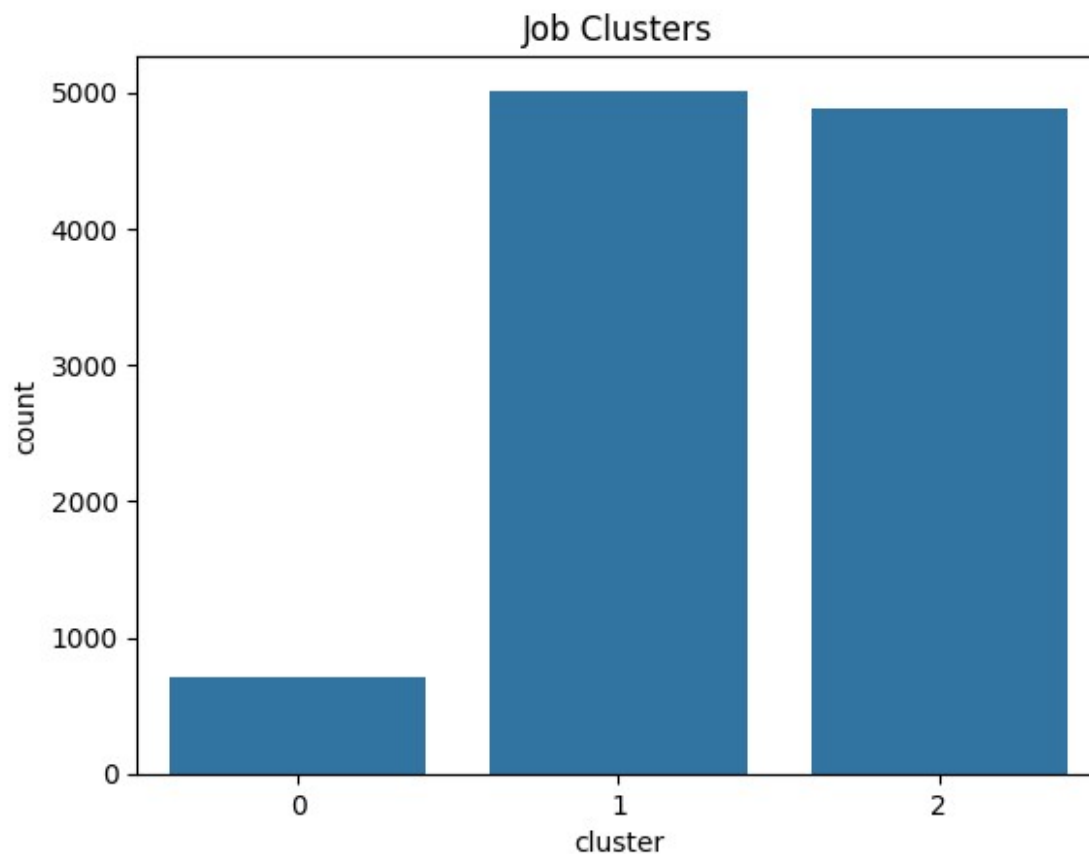
Confusion Matrix:

```
[[1737  297]
 [   39   47]]
```

```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=3, random_state=42)
df['cluster'] = kmeans.fit_predict(X)

sns.countplot(x='cluster', data=df)
plt.title("Job Clusters")
plt.show()
```



```
df.groupby('cluster')['fraudulent'].mean()
cluster
0    0.039773
1    0.036099
2    0.041197
Name: fraudulent, dtype: float64

import pickle

pickle.dump(model, open("model.pkl", "wb"))

import pickle

# Save trained model
pickle.dump(model, open("fraud_model.pkl", "wb"))

from google.colab import files
files.download("fraud_model.pkl")

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

loaded_model = pickle.load(open("fraud_model.pkl", "rb"))
```